Phase 3 AI-Assisted Data Analysis Report

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Dataset: Superstore Sales Transactions (Retail)
Rows / Columns: 9,995 rows including header (9,994 records), 13 columns
Tool Used: Excel (Web) + AI assistant (ChatGPT)
Goal: Explore factors that influence Profit, evaluate relationships using correlation and regression, and identify actionable business insights.

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1. Dataset Overview

This dataset contains retail transaction records with shipping and customer segment fields, location fields (City, State, Region), product grouping fields (Category, Sub-Category), and numeric measures (Sales, Quantity, Discount, Profit).

Key variables used in analysis:

- **Dependent (Outcome):** Profit
- **Predictors (Numeric):** Sales, Quantity, Discount
- **Grouping (Categorical):** Region (Central, East, South, West)

2. Data Cleaning and Preparation

To ensure the analysis was accurate and consistent, the dataset was reviewed for missing values, blanks, and duplicate rows.

Cleaning actions performed:

- Verified the dataset contained 0 blank cells and 0 “NA” values in the working range.
- Identified and removed 17 duplicate rows using Excel’s Remove Duplicates feature across all columns.
- Confirmed data types:
    - Text / categorical: Ship Mode, Segment, Country, City, State, Region, Category, Sub-Category
    - Numeric: Postal Code (treated as categorical identifier), Sales, Quantity, Discount, Profit

Column	Data Type	Continuous/Discrete	Categorical/Numerical	Independent/Dependent	Out of Spec Values	Invalid Data
Ship Mode	Text	Discrete	Categorical		0	0
Segment	Text	Discrete	Categorical		0	0
Country	Text	Discrete	Categorical		0	0
City	Text	Discrete	Categorical		0	0
State	Text	Discrete	Categorical		0	0
Postal Code	Number	Discrete	Categorical		0	0
Region	Text	Discrete	Categorical		0	0
Category	Text	Discrete	Categorical		0	0
Sub-Category	Text	Discrete	Categorical		0	0
Sales	Number	Continuous	Numerical	Independent	0	0
Quantity	Number	Discrete	Numerical	Independent	0	0
Discount	Number	Continuous	Numerical	Independent	0	0
Profit	Number	Continuous	Numerical	Dependent	0	0

Initial observation:
Profit values ranged widely from significant losses to large gains, meaning profitability outcomes vary a lot across transactions.

3. Exploratory Data Analysis (EDA)

Summary statistics showed strong variability, especially for Sales and Profit:

- Sales ranged from very small purchases to very large transactions (high spread).
- Profit included major negative values and major positive values (wide range).

This suggested the presence of extreme outcomes and potential outliers. The dataset was appropriate for outlier detection and further modeling.

4. Outlier Detection (IQR Method)

Outliers were evaluated using the Interquartile Range method:

- IQR = Q3 − Q1
- Effective Minimum = Q1 − 1.5 × IQR
- Effective Maximum = Q3 + 1.5 × IQR
- Outliers = values below Effective Minimum or above Effective Maximum

Outlier counts were calculated for Sales, Quantity, Discount, and Profit. The presence of many outliers confirmed the data is highly skewed, especially in Sales and Profit.

Outliers							
Variable	Q1	Median	Q3	IQR	Eff_Min	Eff_Max	#Outliers
Sales	$ 17.30	$ 54.82	$ 209.97	$ 192.67	$ (271.71)	$ 306.31	1856
Quantity	2	3.00	5	3	-2.5	6.50	1291
Discount	0	0.20	0.2	0.2	-0.3	0.30	1165
Profit	$ 1.73	$ 8.67	$ 29.37	$ 27.65	$ (39.74)	$ 43.19	2517

Interpretation:
Outliers are expected in retail data (bulk purchases, heavy discounts, returns, and special promotions). Instead of deleting them automatically, they were kept and handled through interpretation and visual scaling.

5. Correlation Analysis

A correlation heatmap was created using the numeric fields:

- **Sales**
- **Quantity**
- **Discount**
- **Profit**

Correlation Heatmap				
	Sales	Quantity	Discount	Profit
Sales	1	0.20	-0.03	0.48
Quantity	0.20	1	0.01	0.07
Discount	-0.03	0.01	1	-0.22
Profit	0.48	0.07	-0.22	1

Main results:

- **Profit vs Sales:** 0.48 positive correlation (moderate)
- **Profit vs Discount:** -0.22 negative correlation (moderate)
- **Profit vs Quantity:** 0.07 weak correlation

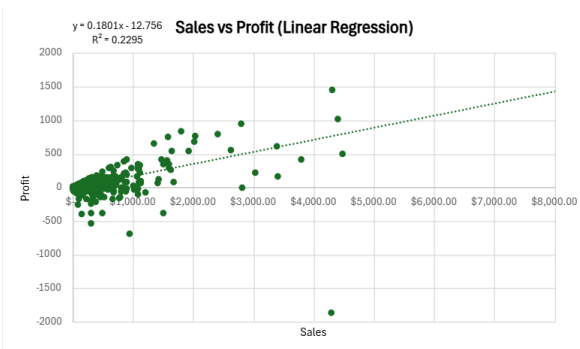
Conclusion:
Sales is associated with higher profit, while discounting tends to reduce profit. However, correlation alone does not explain why profitability differs, so regression and segmentation were used next.

6. Regression Modeling

6.1 Simple Linear Regression (Sales → Profit)

A simple linear regression model was created to predict Profit from Sales:

- Equation shown on chart: y = 0.1801x − 12.756
- R-Squared shown on chart: 0.2295

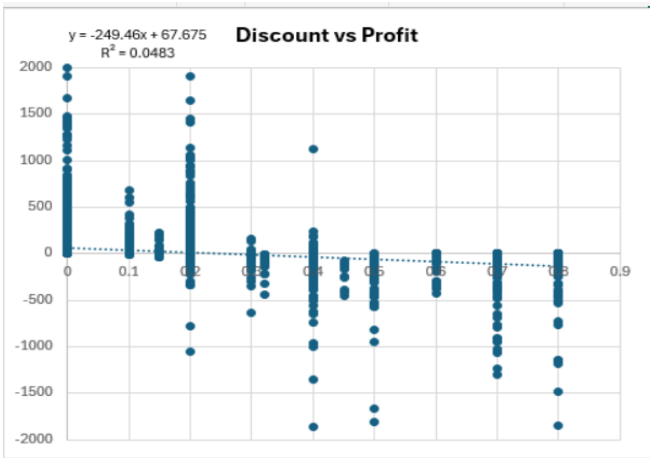


Interpretation:
Sales explains about 23% of the variance in Profit. This confirms Sales matters, but most profit variation is explained by other factors.

6.2 Simple Linear Regression (Discount → Profit)

Discount was tested as a predictor:

- R-Squared was much smaller than the Sales model showing 0.048.
- Discount alone does not strongly predict profit across all transactions.



Interpretation:

Discount has a negative relationship with Profit, but by itself it does not fully explain profit outcomes.

6.3 Multiple Regression (Sales + Discount + Quantity → Profit)

A multiple regression model was created using three predictors:

- Sales
- Discount
- Quantity

The combined model produced:

- R-Squared = 0.2728

Multiple Regression				
Row Label	Quantity	Discount	Sales	Intercept (Profit)
Coefficient	-2.9615	-233.7822	0.1800	35.0216
Std Error	0.9179	9.7024	0.0033	4.2242
Model Fit				
R^2	0.2728		Std Error (Model)	199.9698
F	1246.9277		Observations	9973.0000
Regression SS	149586089.2693		Residual SS	398799407.4795

Interpretation:
Using multiple predictors increases explained variance from 23% to 27%. This is an improvement but still shows that profitability is influenced by additional factors not included in the regression (for example: product mix, region policies, returns, or operational decisions).

7. Model Evaluation (STEYX and RMSE)

Model error was measured using:

- **STEYX** (standard error of estimate)
- **RMSE** (root mean squared error)

RMSE results:

- Sales model RMSE = 205.79
- Discount model RMSE = 228.72

Model Performance Metric (RMSE)	
RMSE (Root Mean Squared Error)	RMSE (Root Mean Squared Error) for Discount
205.79	228.72

STEYX results:

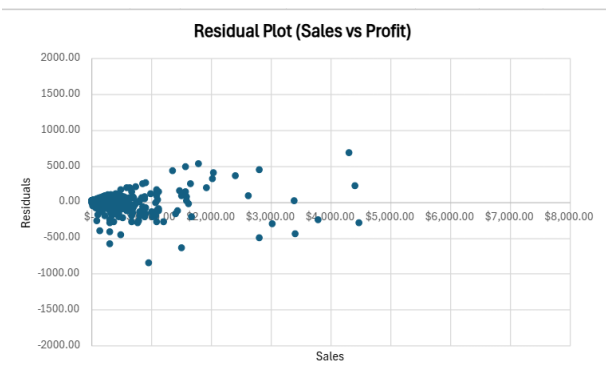
- Sales model STEYX = 205.81
- Discount model STEYX = 228.74
- Formula: =STEYX(Cleaned_Data!M2:M9978, Cleaned_Data!J2:J9978)

Slope			
Sales			
Slope (YX)	Intercept	STEYX	RSQ
0.18	-12.76	205.81	0.23
Discount			
Slope (YX)	Intercept	STEYX	RSQ
-249.4560257	67.67	228.74	0.048

Interpretation:
The Sales-based model produces smaller prediction error than the Discount based model. The lower RMSE confirms that Sales based predictions are more reliable than Discount only predictions.

8. Residual Analysis

A residual plot was created:



Key observation:
Residuals are centered around zero but spread increases slightly at higher sales values. This suggests that the linear model captures the general trend but struggles with high variance transactions and extreme outcomes.

Interpretation:
This supports the idea that profitability is influenced by factors not captured in the simple model, including business decisions and product/location effects.

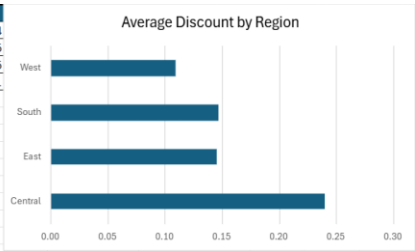
9. Geographic (Region) Insight and Business Findings

Because regression alone did not fully explain Profit, the dataset was segmented by Region to identify patterns that could drive a business decision.

Two visuals were created:

- *Average Profit by Region*
- *Average Discount by Region*

Region	Average Profit	Average Discount
Central	$ 17.10	0.24
East	$ 32.16	0.15
South	$ 28.86	0.15
West	$ 33.93	0.11



Regional pattern observed:

- Central region shows lowest average profit and highest average discount.
- Average sales were relatively similar across regions, meaning the profit gap is not simply explained by sales volume.

Business interpretation:
A likely driver of low profit in the Central region is discount strategy. Heavy discounting may be used to increase volume or hit sales targets, but it appears to reduce profitability.

10. Limitations and Next Steps

Limitations:

- The dataset does not include time fields, promotion dates, or customer-level repeat behavior.
- Profit can be impacted by factors not represented here (returns, shipping costs, product costs, and region-specific pricing policies).
- Extreme outliers influence model fit and visualization scales.

Recommended next steps:

1. Segment further by Category and Sub-Category within each region to see if product mix is driving the Central region's profit problem.
2. Compare profit and discount patterns by Sub-Category to identify "loss leaders" vs healthy margin products.
3. Test a targeted discount reduction strategy: if discount decreases slightly in Central, does profit improve without major sales loss?
4. Build a small dashboard view to allow filtering by Region → Category → Sub-Category to support business decision-making.

11. Conclusion

This analysis combined data cleaning, outlier review, correlation, regression, RMSE error evaluation, and visualization. While regression confirmed that Sales and Discount relate to Profit, the strongest business insight came from segmentation: the Central region shows unusually high discounting and unusually low profit despite similar sales levels.

This creates a clear and actionable direction for a business: investigate discount policy and product mix in Central to improve profitability.